

6

Lines and Angles



In the construction industry, angles make the difference of whether a building is safe or not. Architects and contractors need to calculate angles very precisely to create a structure that stands upright and allows rainwater to run off the roof.



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TOPIC 1

INTRODUCTION TO LINES AND ANGLES

Learning Objectives

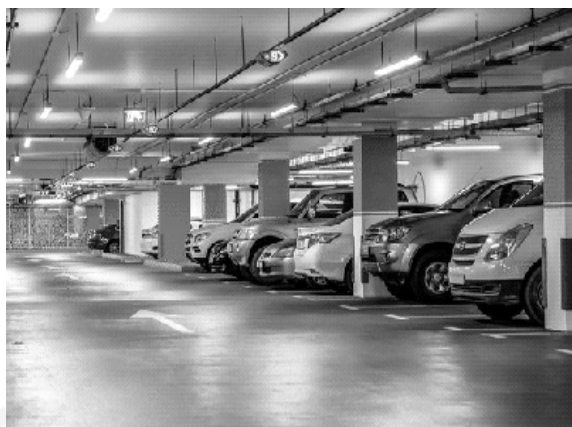
→ Students will be learning the basics of lines and angles.

Learning Outcomes

→ Students will be able to find the angles.

Real Life Application

You must have seen a parking lot in malls or basement area of buildings, where many automobiles are parked at a specific location. There are some markings in that area to avoid confusion and provide a larger area for parking vehicles. The parking areas are thoughtfully planned because it is crucial that automobiles be parked and retrieved in the right way.



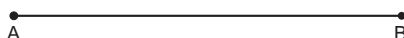
Automobiles are parked in obtuse-angled positions in parking lot because it is easier to turn a car or a two wheeler at an oblique angle than at a right angle, so this may reduce the accidents in a parking lot due to the use of angled spaces.

So, to design such places, designers used the concept of lines and angles. Let us understand the basics of lines and angles.

Basic Definitions and Terms of Lines

Line Segment

A line segment is a part of the line that is bounded by two distinct endpoints. It has a fixed length. In the figure below, shows a line segment and it is denoted by $\overline{AB}$.



Ray

A line which has a fixed starting point but no ending point is called a ray. In the figure given below, it shows a ray and it is denoted by $\overrightarrow{CD}$.



Important

→ Line segments or rays can be parallel or perpendicular.

Collinear Points

Collinear points are the points that lie on the same straight line or in a single line.



Important

→ Collinear points might be close or far away but they will always be in line.



Caution

→ A triangle can only be constructed with a maximum of 2 collinear points if more than 2 collinear points are used, then the construction of a triangle is not possible.

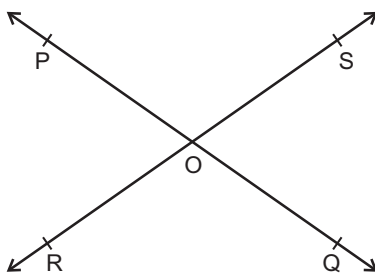
Non-collinear Points

Non-collinear points are defined as three or more points that do not lie on the same line.



Intersecting and Non-intersecting Lines

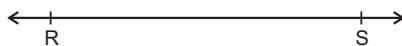
Intersecting lines are those lines that meet or cross each other in a plane from a point.



In the above figure, PQ and RS are the intersecting lines, intersect each other at point O.

Two lines are said to be non-intersecting when the perpendicular distance between them is the same everywhere and they do not meet.



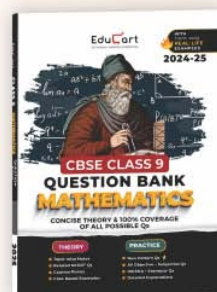


In the above figure, PQ and RS are the non-intersecting lines.



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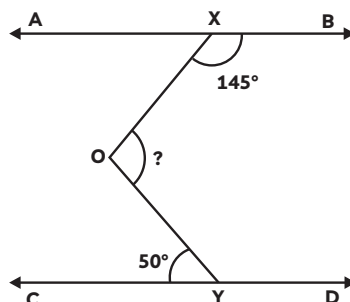


OBJECTIVE Type Questions

[1 mark]

Multiple Choice Questions

1. In the given figure, if $AB \parallel CD$, $\angle OXB = 145^\circ$ and $\angle OYC = 50^\circ$, then $\angle XOY$ is equal to:

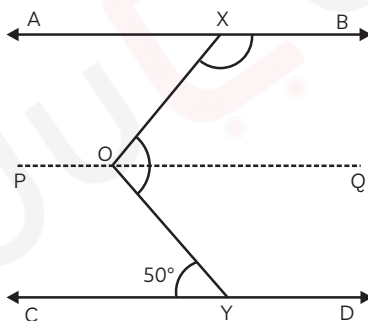


- (a) 65° (b) 75°
(c) 85° (d) 60°

[Diksha]

Ans.(c) 85°

Explanation: Here, we need to draw a line PQ parallel to line AB, through point O as shown in figure.



Note that PQ will also be parallel to CD as lines parallel to the same line are parallel to each other as well.

$$\angle BXO + \angle XOQ = 180^\circ$$

[Since $PQ \parallel AB$ and XO is a transversal, hence Interior angle on the same side of the transversal XO are supplementary]

$$\text{But, } \angle BXO = 145^\circ$$

$$\text{So, } 145^\circ + \angle XOQ = 180^\circ$$

$$\angle XOQ = 180^\circ - 145^\circ$$

$$\angle XOQ = 35^\circ$$

...(i)



Now, $\angle QOY = \angle OYC$

[Since $PQ \parallel CD$ and OY is a transversal, hence alternate Interior angles are equal]

Therefore, $\angle QOY = 50^\circ$... (ii)

Adding eqs. (i) and eqs. (ii)

$$\angle XOQ + \angle QOY = 35^\circ + 50^\circ$$

$$\text{i.e., } \angle XOY = 85^\circ$$

Fill in the Blanks

2. If a transversal line intersects two parallel lines, then a pair of angles is called as

Ans. corresponding angles

Explanation: If a transversal line intersects two parallel lines, then the angles that occupy the same position are called corresponding angles.

True and False

3. Three points will be non-collinear only when they lie on a line.

Ans. False

Explanation: Three points are said to be collinear, if they lie on the same straight line.

Assertion and Reason (A-R)

In these questions, a statement of Assertion (A) is followed by a statement of Reason (R).

Choose the correct option as:

- (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).
 - (b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).
 - (c) Assertion (A) is true but reason (R) is false.
 - (d) Assertion (A) is false but reason (R) is true.
4. Assertion (A): Two angles $a = 30^\circ$ and $b = 50^\circ$ form a pair of complementary angles.

Reason (R): The sum of complementary angles is 90° .

Ans. (d) Assertion (A) is false but reason (R) is true.

Explanation: Here, $a + b = 30^\circ + 50^\circ = 80^\circ \neq 90^\circ$

But, the sum of complementary angles is 90° .

CASE BASED Questions (CBQs)

[4 & 5 marks]

Read the following passages and answer the questions that follow:

5. In a school, the principal of the school decided to celebrate Saraswati Puja and he set the program of the function according to the school clock. On that day, the school timing was 10 am to 8 pm.



- (A) If the first function of pooja starts at 11 am, then find the angle made by hour hand since the school's opening time.
(B) Find the angle made by an hour hand in 50 minutes.
(C) if one of the angles of a linear pair is 75° , then find the other angle.

Ans.(A) Since, a clock has 12 hours.

Therefore, $360^\circ = 12 \text{ hours}$

$$\frac{360^\circ}{12} = 1 \text{ hour}$$

$$1 \text{ hour} = 30^\circ$$

Thus, the angle between 10 am to 11 am made by the hour hand of the clock will be $1 \times 30^\circ = 30^\circ$

(B) We have,

For hour hand,

$$12 \text{ hours} = 360^\circ$$

$$1 \text{ hour} = \frac{360^\circ}{12} = 30^\circ$$

$$60 \text{ minutes} = 30^\circ$$

$$1 \text{ min} = \frac{30^\circ}{60}$$

$$\begin{aligned} \text{Thus, } 50 \text{ minutes} &= \frac{30^\circ}{60} \times 50 \\ &= 25^\circ \end{aligned}$$

(C) Let one angle be $x = 75^\circ$ and other angle be y .

$$x + y = 180^\circ$$



$$75^\circ + y = 180^\circ$$

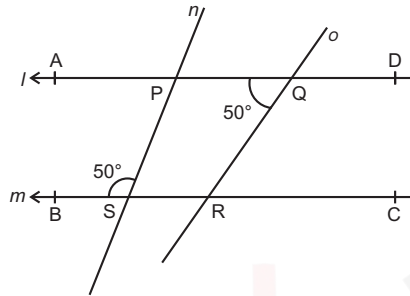
$$y = 180^\circ - 75^\circ$$

$$y = 105^\circ$$

VERY SHORT ANSWER Type Questions (VSA)

[1 mark]

6. In the given figure, l and m are two parallel roads, n and o are two transversals which intersect l and m at point P, S, Q and R respectively. If $\angle PQR = 50^\circ$ and $\angle PSB = 50^\circ$. Is $PS \parallel QR$? If yes, then find $\angle APS$.



Ans. From the given figure,
 $\angle APS + \angle PSB = 180^\circ$ [Co-interior angles]
 $\angle APS + 50^\circ = 180^\circ$
 $\angle APS = 130^\circ$

Here, $\angle APS$ and $\angle PQR$ make a pair of corresponding angles. But, here,
 $\angle APS \neq \angle PQR$

Hence, PS is not parallel to QR .

SHORT ANSWER Type-I Questions (SA-I)

[2 marks]

7. By contributing money, 5 friends bought pizza. They want to divide it equally among themselves. But one of them was given double share, as he was very hungry. Find the angle of the slice of pizza each one received. [Delhi Gov. QB 2022]

Ans. Let, the angles of the friend's pizza be 'x'.

$\Rightarrow$ Each friend got x.

$\Rightarrow$ Hunger friend got '2x'.

Total = 5 friends.

$$\therefore x + x + x + x + 2x = 360^\circ$$

$$\Rightarrow 6x = 360^\circ$$

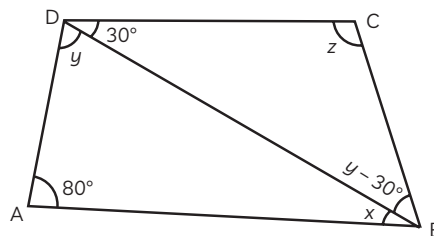
$$\Rightarrow x = 60^\circ$$

Therefore, 4 friends = 60° ; 5th friend = 120°

SHORT ANSWER Type-II Questions (SA-II)

[3 marks]

8. In the given figure if $AB \parallel DC$ and $\angle BDC = 30^\circ$, $\angle BAD = 80^\circ$, find $\angle x$, $\angle y$, $\angle z$.



[Delhi Gov. QB 2022]

Ans. BD is transversal

$$x = 30^\circ$$

[Alternate angle]

$\triangle ABD$,

Sum of angles in triangle = 180°

$$80^\circ + y + x = 180^\circ$$

$$y + 30^\circ = 180^\circ - 80^\circ = 100^\circ$$

$$y = 100^\circ - 30$$

$$y = 70^\circ$$

$\triangle BCD$,

$$\angle BDC = 30^\circ$$

$$\angle BCD = z$$

$$\angle CBD = 70^\circ - 30^\circ = 40^\circ$$

$$\angle BDC + \angle BCD + \angle CBD = 180^\circ$$

[Angle sum property]

$$30^\circ + z + 40 = 180^\circ$$

$$z + 70 = 180^\circ$$

$$z = 180^\circ - 70^\circ$$

$$z = 110^\circ$$

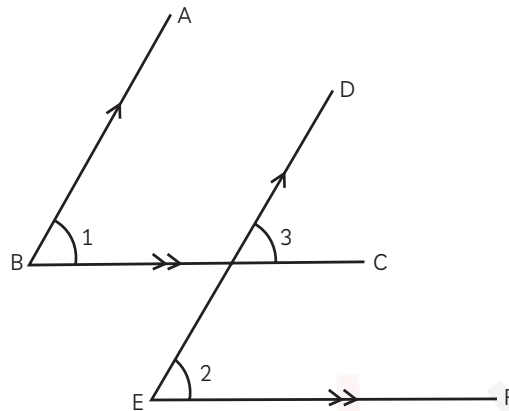


LONG ANSWER Type Questions (LA)

[4 & 5 marks]

9. If the arms of one angle are respectively parallel to the arms of another angle, then show that the two angles are either equal or supplementary.[Delhi Gov. QB 2022]

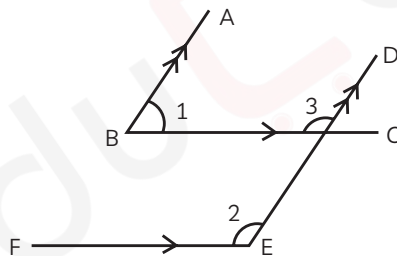
Ans.



$BA \parallel ED$ and BC cuts them.

$$\therefore \angle 1 = \angle 3$$

[Corresponding Angle]



Again, $BC \parallel EF$ and ED cuts.

$$\therefore \angle B = \angle 2$$

[Corresponding Angle]

$$\angle 1 = \angle 2$$

$\therefore$ The Angles are equal

$AB \parallel DE$ and BC cuts.

$$\angle 1 + \angle 3 = 180^\circ$$

[Co-interior Angle]

$BC \parallel EF$ and DE cuts.

$$\therefore \angle 2 = \angle 3$$

[Corresponding Angle]

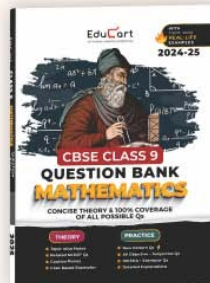
$$\therefore \angle 1 + \angle 2 = 180^\circ$$

[Supplementary]



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